Theorem proving

Direct proof · contrapositive · contradiction · disproof by counterexample

Topics: direct proof, contrapositive, proof by contradiction, and disproof by counterexample.

Today's goal: turn a mathematical statement into a clear argument that shows **why it must be true**.

Session
5 of 15

Date
Fri 11 Sep 2026

COURSE OUTLINE

Where session 5 sits

1 · Course overview and school-maths refresher	2 · Sets and set operations
3 · Functions	4 · Function families
5 · Theorem proving	6 · Mathematical induction
7 · Elementary combinatorics and binomial theorem	8 · Midterm
9 · GCD and divisibility	10 · Modular arithmetic
11 · Vectors	12 · Polynomials
13 · Complex numbers	14 · Course recap and exam preparation
15 · Final exam	

TODAY Direct proof, contrapositive, contradiction, and counterexamples.

WARM-UP

From precise language to proof

In earlier sessions, we learned to name objects precisely:

- → a set has elements;
- → a function has a domain, codomain, and rule;
- → a formula may be true for some inputs and false for others.

Today's question

Once a statement is written clearly, how do we show it is true in every case it claims?

OBJECTIVES

By the end of today, you can...

- → read an "if ... then ..." statement and distinguish it from its converse and contrapositive;
- → write a direct proof with a visible hypothesis and conclusion;
- → use a contrapositive when its starting point is simpler;
- → use contradiction to rule out an impossible assumption;
- → disprove a "for every" statement with one valid counterexample;
- → write the opposite of statements involving "for every" and "there exists."

Success

every line of a proof has a reason.

A theorem is more than a pattern

DEFINITION A **statement** is a sentence that is either true or false.

DEFINITION A **theorem** is a true mathematical statement with a proof.

SENTENCE	STATEMENT?	WHY?
"$x + 1 = 4$"	No	The value of x is not specified.
"For every real x, $x^2 \geq 0$"	Yes	It makes one claim about all real numbers.
"There is an integer n with $n^2 = 9$"	Yes	It says that at least one integer works.

An implication has a hypothesis and a conclusion

DEFINITION An implication has the form

$$P \implies Q,$$

read as "if P, then Q."

- P is the hypothesis (or condition).
- Q is the conclusion.

EXAMPLE If an integer n is a multiple of 4, then n is even.

THE WORDS, IN ALGEBRA Here, "multiple of 4" means $n = 4k$ for some integer k; "even" means $n = 2m$ for some integer m.

Do not confuse an implication with its converse

NOTATION For an implication $P \Rightarrow Q$. Here, $\neg P$ means "not P."

NAME	FORM
original statement	$P \Rightarrow Q$
converse	$Q \Rightarrow P$
contrapositive	$\neg Q \Rightarrow \neg P$

THEOREM The original statement and its contrapositive are always either both true or both false.

EXAMPLE "multiple of $4 \Rightarrow$ even" is true, but its converse is false: 2 is even and is not a multiple of 4.

ACTIVITY 1

Read the logical direction

ACTIVITY 1 PAIRS 10 MIN

Let P be "n is even" and Q be "n^2 is even," where n is an integer.

1. Write the converse of $P \Rightarrow Q$ in words.
2. Write the contrapositive of $P \Rightarrow Q$ in words.
3. For "if n is a multiple of 4, then n is even," test its converse. Is it true or false? Justify your answer.

Share: explain why a counterexample to a converse does not disprove the original implication.

Keep the order when you negate

METHOD For $P \Rightarrow Q$:

1. **Converse:** swap the two parts: $Q \Rightarrow P$.
2. **Contrapositive:** swap the parts **and** negate both: $\neg Q \Rightarrow \neg P$.

REQUESTED STATEMENT	ANSWER
converse of "even $\Rightarrow$ square even"	"if n^2 is even, then n is even"
contrapositive	"if n^2 is odd, then n is odd"
converse of "multiple of $4 \Rightarrow$ even"	false; $n = 2$ is a counterexample

KEY POINT Proving a converse proves a different statement. To prove $P \Rightarrow Q$, prove $P \Rightarrow Q$ itself or prove $\neg Q \Rightarrow \neg P$.

Direct proof: start from the hypothesis

THE SHAPE To prove

$$\text{for every object, if } P \text{ then } Q,$$

use this reliable structure:

METHOD

1. Choose any allowed object that satisfies P.
2. Replace words in P by their definitions.
3. Use algebra or known facts.
4. Reach Q and say why it is true for this object.

NEVER Never assume Q at the start: that is the conclusion you are trying to prove.

The sum of two even integers is even

CLAIM If a and b are even integers, then $a + b$ is even.

PROOF Let a and b be any even integers. Then

$$a = 2r \quad \text{and} \quad b = 2s$$

for some integers r and s. Therefore

$$a + b = 2r + 2s = 2(r + s).$$

Since $r + s$ is an integer, $a + b$ is 2 times an integer. Hence $a + b$ is even. ■

ACTIVITY 2

Write a short direct proof

ACTIVITY 2 PAIRS 12 MIN

Prove the following statement directly:

If a is an even integer and b is an integer, then ab is even.

Use this checklist:

1. Start with any a and b that satisfy the hypothesis.
2. Write $a = 2k$ for an integer k.
3. Rewrite ab in the form $2 \times (\text{an integer})$.

Share: name the expression that must be an integer before the proof is complete.

A direct-proof walkthrough

METHOD

1. Open with the hypothesis, not the conclusion.
2. Substitute the definition into the expression you need to study.
3. Finish by showing that your answer has the form required by the conclusion.

PROOF Let a be an even integer and let b be an integer. Since a is even, $a = 2k$ for some $k \in \mathbb{Z}$. Thus

$$ab = (2k)b = 2(kb).$$

Because $k, b \in \mathbb{Z}$, we have $kb \in \mathbb{Z}$. Therefore ab is even. ∎

THE TRAP $2(kb)$ only proves "even" after we state that kb is an integer.

SECTION 2

When a direct proof is not the clearest route

Sometimes it is hard to work forwards from the hypothesis to the conclusion. Then we change the viewpoint:

· prove the **contrapositive**;

· or assume the claim is false and reach a **contradiction**.

2

Contrapositive: prove the equivalent statement

EQUIVALENCE The statements

$$P \Rightarrow Q \qquad \text{and} \qquad \neg Q \Rightarrow \neg P$$

say the same thing in a different form: they are both true or both false.

They fail only when P is true and Q is false.

METHOD So, to prove $P \Rightarrow Q$ by contraposition:

1. Assume $\neg Q$.
2. Deduce $\neg P$.
3. Conclude that $P \Rightarrow Q$ is true.

WHEN TO USE IT Use this method when "not Q" is easier to work with than P.

If n^2 is even, then n is even

CHANGE THE VIEWPOINT Instead of starting from "n^2 is even," prove the contrapositive:

If n is odd, then n^2 is odd.

PROOF Let n be odd. Then $n = 2k + 1$ for some $k \in \mathbb{Z}$. Hence

$$\begin{aligned} n^2 &= (2k+1)^2 \\ &= 4k^2 + 4k + 1 \\ &= 2(2k^2 + 2k) + 1. \end{aligned}$$

The number $2k^2 + 2k$ is an integer, so n^2 is odd. Therefore, if n^2 is even, then n is even. ∎

Proof by contradiction: an impossible assumption

METHOD To prove a statement P by contradiction:

1. Assume that P is false; that is, assume $\neg P$.
2. Reason carefully from that assumption.
3. Reach a contradiction: for example, a statement and "not that statement," or a fact known to be impossible.
4. Therefore the assumption $\neg P$ was false, so P is true.

THE CONDITION The contradiction must be a consequence of the assumption, not merely a surprising result.

Set up $\sqrt{2}$

CLAIM $\sqrt{2}$ is irrational.

DEFINITION A rational number can be written as p/q, where $p, q \in \mathbb{Z}$ and $q \neq 0$.

Irrational means "not rational."

ASSUME, FOR CONTRADICTION Assume, for contradiction, that

$$\sqrt{2} = \frac{p}{q}$$

where p and q have no common whole-number factor greater than 1 (the fraction is in lowest terms). Squaring gives

$$2q^2 = p^2.$$

Thus p^2 is even. By the previous theorem, p is even.

Finish $\sqrt{2}$

PROOF, CONTINUED Since p is even, write $p = 2k$ for some $k \in \mathbb{Z}$. From $2q^2 = p^2$,

$$2q^2 = (2k)^2 = 4k^2,$$

so

$$q^2 = 2k^2.$$

Thus q^2 is even, and therefore q is even as well.

Both p and q are divisible by 2. This contradicts that p/q was in lowest terms. Therefore $\sqrt{2}$ is irrational. ∎

THE CONTRADICTION "No common factor greater than 1" conflicts with "both p and q are divisible by 2."

ACTIVITY 3

Choose a proof method before writing

ACTIVITY 3 GROUPS OF THREE 12 MIN

For each claim, choose the proof method that makes the starting condition easiest to use. Then write the first two lines.

CLAIM	YOUR METHOD
If a and b are odd integers, then $a+b$ is even.	?
If n^2 is odd, then n is odd.	?
$\sqrt{2}$ is irrational.	?

Share: one group explains the method choice, not only the first calculation.

Why these methods fit

CLAIM	GOOD WAY TO START
odd a, odd $b \Rightarrow a + b$ even	Direct: let $a = 2r + 1$ and $b = 2s + 1$.
n^2 odd $\Rightarrow n$ odd	Contrapositive: assume n is even, so $n = 2k$ and $n^2 = 2(2k^2)$ is even.
$\sqrt{2}$ irrational	Contradiction: assume $\sqrt{2} = p/q$ in lowest terms.

METHOD **Choosing a method is part of writing a proof.** Ask: which assumption gives me algebra I can use straight away?

A counterexample can disprove a "for every" claim

DEFINITION To disprove a statement that says

$$\text{"for every } x \in D,\ P(x)\text{"},$$

find one allowed value $a \in D$ for which $P(a)$ is false. Here, D is the set of allowed values.

EXAMPLES

FALSE "FOR EVERY" STATEMENT	COUNTEREXAMPLE
Every prime number is odd.	2 is prime and even.
For every real x, $x^2 \geq x$.	At $x = \frac{1}{2}$, $x^2 = \frac{1}{4} < \frac{1}{2} = x$.

BOTH DIRECTIONS One example that works does **not** prove a statement about every value. To disprove a statement that says "there exists," show that no allowed value works.

ACTIVITY 4

Disprove, prove, or explain why one example is not enough

ACTIVITY 4 PAIRS 12 MIN

For each statement, decide whether it is true or false. Give the requested justification.

1. For every real x, $x^2 \geq 0$. If true, state the known fact you use.
2. For every integer n, $n^2 + n$ is odd. If false, give a counterexample.
3. For every real x, $(x+1)^2 = x^2 + 1$. If false, give a counterexample.
4. There exists a real x with $x^2 = -1$. Explain why checking a few values is not a disproof.

Share: give one valid counterexample and name the statement type it disproves.

Check "every" or "there exists" first

ITEM	REASONING
1	True. For every real x, the square x^2 is non-negative.
2	False. At $n = 1$, $n^2 + n = 2$, which is even.
3	False. At $x = 1$, the left side is 4 and the right side is 2.
4	False. Since every real square is at least 0, no real square can equal -1.

METHOD For item 4, a single failed trial is not enough: the claim says **there exists** a value. We disprove it by showing that **every** real value fails.

Four common proof mistakes

PITFALL

1. Proving $Q \Rightarrow P$ when the question asks for $P \Rightarrow Q$.
2. Assuming the conclusion.
3. Negating "every" or "there exists" incorrectly.
4. Giving a few examples as a proof for all values.

FIX

1. Label the hypothesis and conclusion before you start.
2. Begin with P, or with $\neg Q$ in a contrapositive proof.
3. "Not every $x \in D$ has $P(x)$" means that at least one $x \in D$ makes $P(x)$ false. "No $x \in D$ has $P(x)$" means $P(x)$ is false for every $x \in D$.
4. Examples test an idea; a proof must cover every case.

EXIT TICKET

Work independently

1 Prove directly: if m and n are odd integers, then $m + n$ is even.

2 Prove by contraposition: if n^2 is even, then n is even.

3 Disprove: for every real x, $x^2 + 1 > 2x$.

4 Write the opposite statement: "For every real x, $x^2 > x$."

RULE Write the method name and at least one sentence explaining why your proof or counterexample works.

WRAP-UP

What today was about

- An implication has a **hypothesis** and a **conclusion**. Its converse is a different statement; its contrapositive is true in exactly the same cases.
- A **direct proof** starts from the hypothesis and reaches the conclusion.
- A **contradiction proof** assumes the claim is false and shows that assumption is impossible.
- One valid **counterexample** disproves a statement that claims something is true for every allowed value.

implication · converse · contrapositive · negation · quantifier · contradiction · counterexample

Problem Set 5

Eight statements to prove or disprove. Name the method at the top of each solution and explain each key step.

Next session: **mathematical induction** — a method for proving a statement for infinitely many integers.